

Problem Statement

Evaluate the following limit from the image "dailyintegral-limit-limit-1x1-2026-07-22.jpg":

Let $A(n)$ be the first-quadrant area bounded by the coordinate axes and $x^n + y^n = 1$. Compute:

$$\lim_{k \rightarrow \infty} \prod_{n=1}^k [A(2^n)]^{2^n}$$

Detailed Solution

To solve this limit, we first need an explicit formula for the area $A(n)$. We will express this area using the Beta and Gamma functions, simplify the resulting product to find a telescoping pattern, and finally evaluate the limit using L'Hôpital's rule and the Digamma function.

Step 1: Express $A(n)$ using the Beta function

The area in the first quadrant bounded by $x^n + y^n = 1$ can be set up as an integral by solving for y :

$$y = (1 - x^n)^{\frac{1}{n}}$$

Thus, the area $A(n)$ is given by:

$$A(n) = \int_0^1 (1 - x^n)^{\frac{1}{n}} dx$$

We can transform this integral into the standard form of the Beta function. Let $u = x^n$. Then $x = u^{\frac{1}{n}}$ and $dx = \frac{1}{n} u^{\frac{1}{n}-1} du$. The limits of integration remain 0 and 1. Substituting this into the integral yields:

$$A(n) = \int_0^1 (1 - u)^{\frac{1}{n}} \cdot \frac{1}{n} u^{\frac{1}{n}-1} du = \frac{1}{n} \int_0^1 u^{\frac{1}{n}-1} (1 - u)^{\frac{1}{n}} du$$

By the definition of the Beta function, $B(x, y) = \int_0^1 t^{x-1} (1 - t)^{y-1} dt$, we have:

$$A(n) = \frac{1}{n} B\left(\frac{1}{n}, 1 + \frac{1}{n}\right)$$

Step 2: Simplify $A(n)$ using the Gamma function

We can rewrite the Beta function in terms of the Gamma function, using the identity $B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$:

$$A(n) = \frac{1}{n} \frac{\Gamma\left(\frac{1}{n}\right) \Gamma\left(1 + \frac{1}{n}\right)}{\Gamma\left(1 + \frac{2}{n}\right)}$$

Using the fundamental functional equation of the Gamma function, $\Gamma(z+1) = z\Gamma(z)$, we can write $\Gamma\left(\frac{1}{n}\right) = n\Gamma\left(1 + \frac{1}{n}\right)$. Substituting this back into our expression for $A(n)$:

$$A(n) = \frac{1}{n} \frac{n\Gamma\left(1 + \frac{1}{n}\right)\Gamma\left(1 + \frac{1}{n}\right)}{\Gamma\left(1 + \frac{2}{n}\right)} = \frac{\Gamma\left(1 + \frac{1}{n}\right)^2}{\Gamma\left(1 + \frac{2}{n}\right)}$$

Step 3: Evaluate the telescoping product

We are tasked with evaluating the product $\prod_{n=1}^k [A(2^n)]^{2^n}$. Let's substitute 2^n for n in our simplified area formula:

$$A(2^n) = \frac{\Gamma\left(1 + \frac{1}{2^n}\right)^2}{\Gamma\left(1 + \frac{2}{2^n}\right)} = \frac{\Gamma\left(1 + \frac{1}{2^n}\right)^2}{\Gamma\left(1 + \frac{1}{2^{n-1}}\right)}$$

Now, raise $A(2^n)$ to the power of 2^n :

$$[A(2^n)]^{2^n} = \frac{\Gamma\left(1 + \frac{1}{2^n}\right)^{2^{n+1}}}{\Gamma\left(1 + \frac{1}{2^{n-1}}\right)^{2^n}}$$

Notice that the product is perfectly telescoping. Let P_k be the partial product up to k :

$$P_k = \prod_{n=1}^k \frac{\Gamma\left(1 + \frac{1}{2^n}\right)^{2^{n+1}}}{\Gamma\left(1 + \frac{1}{2^{n-1}}\right)^{2^n}}$$

Writing out the first few terms:

$$P_k = \left(\frac{\Gamma\left(1 + \frac{1}{2}\right)^4}{\Gamma(2)^2}\right) \cdot \left(\frac{\Gamma\left(1 + \frac{1}{4}\right)^8}{\Gamma\left(1 + \frac{1}{2}\right)^4}\right) \cdots \left(\frac{\Gamma\left(1 + \frac{1}{2^k}\right)^{2^{k+1}}}{\Gamma\left(1 + \frac{1}{2^{k-1}}\right)^{2^k}}\right)$$

All intermediate terms cancel out, leaving only the numerator of the final term and the denominator of the first term:

$$P_k = \frac{\Gamma\left(1 + \frac{1}{2^k}\right)^{2^{k+1}}}{\Gamma(2)^2}$$

Since $\Gamma(2) = 1! = 1$, the product simplifies beautifully to:

$$P_k = \Gamma\left(1 + \frac{1}{2^k}\right)^{2^{k+1}}$$

Step 4: Evaluate the limit as $k \rightarrow \infty$

We now need to find $\lim_{k \rightarrow \infty} P_k$. Let $x = \frac{1}{2^k}$. As $k \rightarrow \infty$, $x \rightarrow 0^+$. The limit transforms to:

$$L = \lim_{x \rightarrow 0^+} \Gamma(1+x)^{\frac{2}{x}}$$

To evaluate this, take the natural logarithm of both sides:

$$\ln(L) = \lim_{x \rightarrow 0^+} \frac{2 \ln(\Gamma(1+x))}{x}$$

This is a $\frac{0}{0}$ indeterminate form, so we can apply L'Hôpital's rule. The derivative of $\ln(\Gamma(1+x))$ with respect to x is the Digamma function, $\psi(1+x)$:

$$\ln(L) = 2 \lim_{x \rightarrow 0^+} \psi(1+x) = 2\psi(1)$$

It is a known property of the Digamma function that $\psi(1) = -\gamma$, where γ is the Euler-Mascheroni constant ($\gamma \approx 0.57721$). Thus:

$$\ln(L) = -2\gamma \implies L = e^{-2\gamma}$$

Final Answer:

$$\mathbf{e^{-2\gamma}}$$